E-Cigarette Supply: A Conference-Version Model

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Draft

Motivating facts, model environment, and equilibrium definition for the internal-conference draft. Static pricing, exogenous state, explicit fringe.

1 Motivating facts

The tracked e-cigarette market is small-N and its leadership turns over completely. Figure 1 plots annual, mL-weighted brand composition from the Nielsen RMS panel: the top-5 brands each year, named and individually shaded, with the market leader starred, and everyone else pooled into "Other."

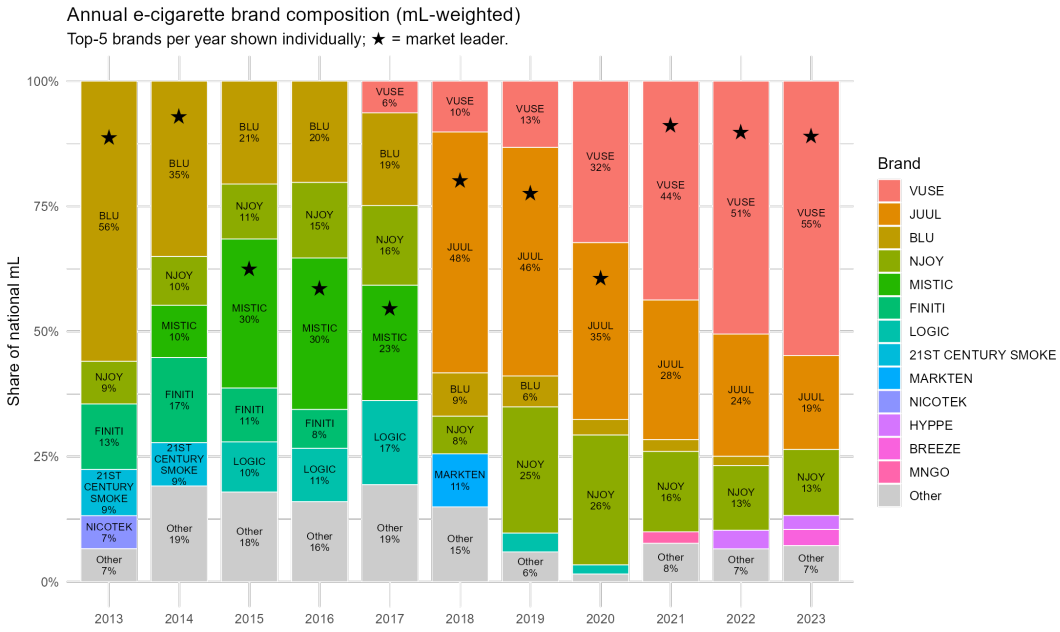


Figure 1: Annual e-cigarette brand composition, mL-weighted, 2013–2023. Source: Nielsen RMS retail scanner panel (national, annual, mL-weighted). Top-5 brands per year shown individually; ⋆ = market leader (CR1). Rest = "Other."

Three things in this figure motivate the model form directly:

1. **A small number of firms account for most of the market, every year.** CR_3 never drops below 55.5% (2015) and reaches 87.9% (2022) — a handful of firms, not a competitive fringe of many, set the terms of competition in any given period. This motivates capping the dynamic game at $\bar{N}$ active firms rather than modeling every brand as a strategic player.

2. **The identity of the leader changes completely, more than once.** BLU (2013) → MISTIC (2015–2017) → JUUL (2018–2020) → VUSE (2021–2023) — three full leadership transitions in an eleven-year panel. A static entry-barriers story doesn't produce this; a dynamic model with state-dependent entry, exit, and (eventually) investment does.

3. **The residual below CR_3 is not a fixed constant.** It ranges from 12–22% of national mL in most years down to 12.3% in 2020 — small but persistent, and volatile enough that it needs to be modeled explicitly rather than assumed away (Table 1; see also the firm-count/materiality diagnostics in the companion `08_27_2026` note, which show the sub-CR_3 group holding 15–22% of national mL through 2014–2018 specifically).

Table 1: Annual concentration, mL-weighted national brand shares.

Year	Leader	Leader share	Runner-up	CR_3	HHI
2013	BLU	56.0%	FINITI	78.3%	3519
2014	BLU	35.0%	FINITI	61.8%	1869
2015	MISTIC	24.0%	BLU	55.5%	1430
2016	MISTIC	25.9%	BLU	61.3%	1564
2017	MISTIC	20.8%	BLU	56.0%	1430
2018	JUUL	46.9%	VUSE	67.7%	2612
2019	JUUL	42.3%	NJOY	77.0%	2609
2020	JUUL	33.7%	VUSE	87.3%	2639
2021	VUSE	43.7%	JUUL	87.7%	2966
2022	VUSE	50.5%	JUUL	87.9%	3338
2023	VUSE	54.8%	JUUL	86.8%	3553

HHI is U-shaped over the panel — highly concentrated in 2013 (∼3,500), falls to the DOJ's moderately-concentrated floor by 2015–2017 (∼1,400–1,600) as cigalike incumbents fragment, then re-concentrates from 2018 on as first JUUL and then VUSE consolidate the market. That U-shape is itself evidence against a static model: concentration is an equilibrium outcome of entry, exit, and (later) investment responding to shocks — first the salt-nicotine innovation, then a wave of FDA actions in 2019–2020 — not a fixed structural feature of the industry.

2 Environment

Players and timing, period t (monthly). Up to $\bar{N} = 5$ incumbent firms are active, indexed $i = 1, \ldots, n_t \leq \bar{N}$, plus one explicit competitive fringe (a single aggregated player, not $\bar{N}$-capped) and at most one potential entrant. Each period: (i) states are realized; (ii) active firms and the fringe compete in a static pricing subgame; (iii) profits realize; (iv) incumbents decide continue/exit and the potential entrant decides enter/stay out; (v) states evolve into $t + 1$.

State. Firm i carries $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$ — an accumulated-quality index and an idiosyncratic residual shock, both firm-and-time-indexed. In this version both are **exogenous**, evolving as first-order Markov processes estimated directly off the data:

$$a_{i,t+1} \sim F_a(\cdot \mid a_{i,t}), \qquad \xi_{i,t+1} \sim F_\xi(\cdot \mid \xi_{i,t}).$$

The fringe carries a single aggregate state $a_{F,t}$, exogenous, estimated the same way off the fringe's observed aggregate series (net of the 2021 Nielsen-panel-onboarding correction — see the companion model note, §4.6).

3 Demand

A firm-level nested logit: every active firm's product and the fringe sit in one "e-cigarette" nest against a single outside-option nest. Firm i's mean utility is

$$\delta_{i,t} = \beta_a \, a_{i,t} + \xi_{i,t} - \alpha \, p_{i,t},$$

where β_a is a free coefficient on the quality state (not normalized to 1 — $a_{i,t}$ is an observable, physically-scaled quantity, and the Ericson–Pakes (1994) canonical single-period profit function gives its own quality term a free coefficient rather than unit weight) and $\xi_{i,t}$ is the exogenous residual already carried in the state $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$, included per standard mean-utility convention (Berry 1994/BLP). Market shares follow the standard nested-logit form with nest parameter $\sigma \in [0, 1)$ governing within-nest substitution. (Nest correlation σ, the price coefficient α, and β_a are estimated objects, not calibrated here.)

Price and product construction (retail rebuild, 7 October 2026). UPC price $p_{u,t}$ = real revenue / corrected mL sold; firm price is its unweighted mean over eligible UPCs. Positive sales, at least three stores, UNKNOWN excluded. Reviewed pack counts and capacities update volume, nicotine quantities and shares; NCP inputs stay unchanged.

For N UPCs: $\bar{p} = (1/N)\,\Sigma_u\, p_u$, $Q = \Sigma_u\, q_u$, $R = \Sigma_u\, p_u q_u$.
$\bar{p}Q - R = \Sigma_u(\bar{p} - p_u)q_u$, generally nonzero. R/Q recovers observed revenue; the unweighted price need not.

Demand diagnostics also use unweighted cartridge/disposable mL and log eligible UPC count. Incomplete size firm-months are excluded and reported. See the current comparison report for estimates and market-size variants.

Instruments and empirical specification. Price and $\log(s_{i,t}/s_{g,t})$ are jointly instrumented by tax and an active-brand count. Tax is population-weighted over states where firm i sells: $\Sigma_{s \text{ in } S_it}\, [Pop_{s,t}/\Sigma_{s' \text{ in } S_it}\, Pop_{s',t}]\, tax_{s,t}$.

Compare the national all-brand count with the number of distinct brands operating in the union of firm i's states. Both include the focal brand and fringe brands. The empirical incumbent roster uses annual volume share >7%. Specifications compare no firm FE with firm FE, use no month FE, and cluster errors by month. Relevance and exclusion require separate assessment; admissible nesting is not imposed. Empirical tables report the signed price slope; the theoretical equations here retain -αp.

4 Static pricing subgame

Given the period's active-firm set and states, each firm sets price to maximize static profit:

$$\pi_i(\omega_t) \;=\; \max_{p_i} \;\big(p_i - c(\omega_{i,t})\big)\, q_i(p_t, \omega_t) \;-\; FC, \qquad q_i(p_t, \omega_t) \;=\; M_t \cdot s_i(p_t, \omega_t),$$

where q_i is quantity and s_i is the nested-logit market share from §3; market size M_t cancels out of the resulting first-order condition, so the equilibrium price depends only on shares. The Bertrand–Nash equilibrium price vector solves the resulting system of first-order conditions simultaneously across all active firms (and the fringe, if it prices rather than holding a reduced-form price) — a standard logit markup condition, expanded and solved numerically in the companion solution note. **This is genuinely static**: p_t^* depends only on the current-period state ω_t, not on any continuation value. "Price" varies dynamically across the paper's periods only because the state ω_t on which it is conditioned is itself evolving — not because pricing today is chosen with tomorrow in mind.

5 Dynamic entry and exit

Let $\pi_i^*(\omega_t)$ denote the equilibrium static profit from §4. An incumbent's value function:

$$V_i(\omega_{i,t}, \omega_{-i,t}) \;=\; \pi_i^*(\omega_t) \;+\; \max\Big\{ \underbrace{\phi_i}_{\text{exit: scrap value}} \;,\; \underbrace{\beta \cdot \mathbb{E}\big[\, V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue} \,\big]}_{\text{continue}} \Big\}.$$

Optimal exit rule: firm i exits iff $\phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]$. Define the implied exit indicator:

$$\lambda^{\text{exit}}(\omega_{i,t}; \omega_{-i,t}) \;=\; \mathbf{1}\big\{\, \phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}] \,\big\}.$$

The potential entrant compares its expected discounted value of entering against a sunk entry cost x_e; it enters iff the former net of the latter is non-negative. Define the implied entry indicator:

$$\lambda^{\text{entry}}(\omega_t) \;=\; \mathbf{1}\big\{\, \beta \cdot \mathbb{E}[V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t] - x_e \geq 0 \,\big\}.$$

(λ, not η — in Ericson–Pakes' own model η' already denotes something unrelated, an aggregate industry-wide shock inside the transition kernel, and χ is claimed too, as the exit-decision indicator itself in the closely-related Doraszelski–Pakes–Satterthwaite restatement of this model. λ sidesteps both of those as well as the nested-logit correlation parameter σ already used in §3.) Equilibrium is a fixed point in $(\lambda^{\text{exit}}, \lambda^{\text{entry}})$ and the value function, computed via the standard Pakes–McGuire (1994) iteration — full algorithm, plus the transition-probability mechanics these indicators feed into, in the companion solution note.

Does an incumbent factor in a rival's possible exit, or a new entrant? Yes, and this is already fully active in the present version, even before investment becomes endogenous. The value function $V_i(\omega_{i,t}, \omega_{-i,t})$ explicitly takes rivals' states $\omega_{-i,t}$ as an argument, and the continuation value $\mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]$ integrates over the full future state vector, which includes what rivals do. Since ϕ_j and x_e are private draws, firm i cannot know for certain whether rival j will exit or whether the entrant will show up — but it knows the equilibrium cutoff rules (λ^{exit} exit iff ϕ_j exceeds some threshold given j's state; λ^{entry} enter iff expected value exceeds x_e), so it can compute the probability of each, and its own continuation value is an expectation over all of that. "Equilibrium is a fixed point in $(\lambda^{\text{exit}}, \lambda^{\text{entry}})$ and the value function" above is exactly this mutual consistency — every firm's best response already correctly anticipates every other firm's probabilistic policy.

Exit sample note. MarkTen's and MISTIC's exits track dated parent-company decisions (Altria discontinuing MarkTen in 2018; Ballantyne winding MISTIC down ∼2020) rather than the state-dependent comparison above, and are carved out as exogenous ownership shocks; the remaining top-5 exits are the identifying variation for ϕ_i.

6 The fringe

The fringe enters the demand system as one more alternative inside the nest, $\delta_{F,t} = \beta_a \, a_{F,t}$ (the same demand coefficient β_a as the $\bar{N} = 5$ firms; the fringe carries one aggregate state, not a firm-and-time-indexed pair, so there is no separate $\xi_{F,t}$), competing for share against the $\bar{N} = 5$ firms and the outside option in the same static pricing solve. It carries no Bellman equation in this version — no entry/exit decision of its own, just a correctly time-varying aggregate presence. This is the cheap version, chosen specifically to avoid adding dimensions back into the dynamic game's state space right after the state-space size was the reason to move to Experience-Based Equilibrium in the first place (§8).

7 What this version leaves for later

This is a conference draft, not the paper's final specification. Explicitly deferred:

- **Investment.** $a_{i,t}$ is exogenous here. The full model endogenizes it via a binary invest/don't-invest choice ($d_{i,t} \in \{0, 1\}$, a firm-and-time-varying investment cost $\kappa_{\text{inv},i,t}$ if investing, stochastic success), solved as an Experience-Based Equilibrium rather than a full Markov Perfect Equilibrium because the state space (up to 5 firms × 2 states each) is too large to solve directly — per Ariel Pakes's recommendation. $\kappa_{\text{inv},i,t}$ is also where FDA regulation

enters the full model (stepping up at known enforcement dates, with an additional JUUL-specific increment) — chosen over directly capping $a_{i,t}$ because it lets rivals' continuation values respond to a regulation-weakened competitor, which a static/capped-state version cannot do.

- **The cost channel for $a_{i,t}$.** Dropped. Firm-level learning-curve regressions find no price-identified cost-reduction signature for the pod-era leaders (JUUL/VUSE/NJOY) — only for the 2013–2017 cigalike cohort — and price alone can't separate cost movements from markup movements without additional data (margins, compliance costs) this draft doesn't have. $a_{i,t}$ is carried here purely as a demand-side (mean-utility) state.

- **Question : does crossing the incumbent-share threshold count as entry/exit?** Demand-side incumbent status is now defined by annual market share rather than the $\bar{N} = 5$ cap directly: a firm is treated as a full incumbent in the nested logit for a given calendar year iff its annual market share exceeds 7% (the logit itself stays at monthly frequency; only the incumbent/fringe assignment is fixed per year), with firms below that line folded into the fringe aggregate for that year. This is now decoupled from $\bar{N} = 5$, which governs only the dynamic entry/exit game. Open question: when a firm crosses the threshold — in either direction — does that reclassification itself count as dynamic entry or exit (i.e., does it trigger the $\lambda^{\text{exit}}/\lambda^{\text{entry}}$ comparison against ϕ_i/x_e), or is entry/exit reserved for a firm genuinely starting or stopping sales altogether, independent of this share-threshold reclassification? The two interpretations imply different identification strategies for ϕ_i and x_e. At 7%, this is moot in the current data: every firm has a single unbroken incumbent spell (NJOY included, continuous 2013–2023), and $\bar{N} = 5$ never binds either — the annual incumbent count ranges 3–6, hitting 6 in 2014 and again in 2018 as the cigalike-era leaders and the pod-era entrants briefly overlap above the line. **Note: NJOY becomes an issue at a higher threshold.** At 10% or above, NJOY drops below the cutoff in 2018 (7.6% share) and returns from 2019 on — a genuine gap-and-return case, so the entry/exit interpretation question stops being moot and gets entangled with the choice of cutoff itself.

- **A strategic fringe.** The fringe's aggregate state is exogenous here. A richer version replaces it with a monopolistically-competitive free-entry sector whose size responds endogenously to the same shocks hitting the top firms.

- **Question : legacy-brand bias in $a_{F,t}$.** $a_{F,t}$ is an unweighted average of nicotine content across active fringe UPCs each month. Diagnostics find that once a tracked firm permanently exits the top-5 roster, its residual UPCs can systematically bias $a_{F,t}$ if that firm's own nicotine level is a persistent outlier relative to the rest of the fringe — confirmed for two firms (MarkTen, pulling $a_{F,t}$ up ∼12–20% through 2019; NICOTEK, pulling it down ∼10–11%), with every other tracked firm's post-exit residual found negligible. A general correction is derived and validated — exclude a firm's post-exit UPCs from the fringe average once its deviation from that month's fringe mean exceeds ∼10%, applied only to months after a firm's last incumbent year so it never touches a firm's pre-incumbency rise (this distinction matters: a naive version of the rule, applied without the pre/post split, would have wrongly flagged VUSE's and JUUL's own early, genuinely-rising histories) — but not yet implemented. Open question: is this worth correcting in the construction, or is the resulting bias small enough relative to everything else in the model to leave as a documented limitation?

- **What "cost of nicotine" is a cost of, functionally**, and the ownership-vs-quality question as a driver of turnover (§7.6 of the companion diagnostics note) — both flagged, neither resolved, neither needed for this draft.

8 Framework and references

Ericson & Pakes (1995) for the entry/exit/investment framework; Pakes & McGuire (1994, 2001) for the computational algorithm (the second paper's recurrent-states restriction is the direct precursor to the EBE move); Spence (1981) for the learning-by-doing idea originally considered for $a_{i,t}$'s state interpretation (dropped — see §7), with Benkard (2000, 2004) as the specific aircraft-manufacturing application of that idea, not its origin; Fershtman & Pakes (2012) for Experience-Based Equilibrium. Citations characterized from memory at the level of what each paper does — worth confirming exact notation against the originals before this goes into a submitted draft.